

MUNSTER OSNABRUCK, Germany

WMO: 103150

Lat: 52.1344N

Lon: 7.6969E

Elev: 173

StdP: 14.60

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.8	22.6	11.8	10.1	21.7	15.9	12.4	23.9	27.8	47.6	24.8	46.7	5.7	70	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	87.6	67.1	82.7	65.4	79.0	64.0	69.3	81.0	67.6	78.7	66.1	76.3	7.6	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	96.5	72.7	64.2	90.5	71.2	62.6	85.3	69.7	33.4	80.9	32.0	79.3	30.9	76.5	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.1	18.5	16.3	DB	13.5	95.0	5.8	3.2	9.3	97.3	5.9	99.2	2.7	101.0	-1.5	103.4
			WB	12.9	72.8	5.8	1.5	8.7	73.9	5.3	74.8	2.0	75.6	-2.2	76.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.1	36.4	36.9	42.7	50.9	56.6	62.7	66.6	64.5	59.2	52.1	44.1	39.4	(6)
	DBStd	12.52	7.81	7.18	7.40	7.16	7.22	6.25	6.30	5.47	5.63	6.38	6.84	7.72	(7)
	HDD50	1714	424	366	242	74	19	1	0	0	2	52	200	334	(8)
	HDD65	5329	888	786	691	425	277	117	55	76	191	400	628	795	(9)
	CDD50	2106	1	0	16	101	222	381	514	450	278	117	22	4	(10)
	CDD65	248	0	0	0	2	16	47	104	60	18	1	0	0	(11)
	CDH74	2328	0	0	0	41	158	432	1021	520	146	10	0	0	(12)
	CDH80	720	0	0	0	3	31	114	374	166	31	1	0	0	(13)
(14) Wind	WSAvg	7.3	8.7	7.9	8.1	7.1	7.1	6.7	6.6	6.3	6.3	6.6	7.4	8.7	(14)
(15) Precipitation	PrecAvg	30.8	2.8	2.2	2.0	1.8	2.2	2.7	3.1	3.1	2.6	2.5	2.8	2.9	(15)
	PrecMax	40.2	5.3	4.8	5.0	3.9	5.1	7.6	6.4	10.1	5.7	7.5	5.3	6.7	(16)
	PrecMin	23.6	0.2	0.6	0.4	0.1	0.9	0.5	1.1	0.9	0.4	0.6	0.0	1.1	(17)
	PrecStd	4.6	1.3	1.2	1.2	0.8	1.0	1.5	1.4	2.0	1.4	1.3	1.3	1.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	57.2	66.6	79.0	84.3	88.2	94.7	91.6	84.4	74.9	62.9	57.2	(19)
		MCWB	51.2	46.3	52.3	58.8	65.3	67.6	68.7	68.9	65.6	62.3	56.5	52.4	(20)
	2%	DB	52.1	51.8	61.2	73.4	78.4	82.7	88.2	84.3	78.4	68.3	59.0	53.9	(21)
		MCWB	49.0	47.2	50.7	56.4	61.3	66.0	67.1	66.9	64.4	60.6	54.8	50.2	(22)
	5%	DB	49.7	49.7	57.3	69.4	73.8	78.9	84.1	79.0	73.5	64.4	55.8	51.8	(23)
		MCWB	47.4	46.4	49.6	55.8	59.0	64.2	65.9	64.7	62.7	58.6	52.3	48.7	(24)
	10%	DB	46.7	46.7	53.7	64.3	70.0	75.1	79.2	75.4	69.5	61.1	53.9	48.6	(25)
		MCWB	44.8	44.0	47.6	53.6	58.4	63.0	64.6	63.3	61.0	56.7	50.6	46.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.3	50.1	55.1	60.4	67.9	70.7	71.6	70.8	68.1	63.3	58.0	53.2	(27)
		MCDB	54.6	51.7	62.1	73.6	81.5	82.9	85.4	82.5	79.4	70.9	61.0	56.1	(28)
	2%	WB	49.8	48.1	52.5	58.3	64.2	68.4	69.3	68.5	65.7	61.0	55.3	51.2	(29)
		MCDB	52.2	50.7	58.8	69.7	74.3	79.1	81.6	80.4	75.7	67.5	58.6	53.9	(30)
	5%	WB	47.2	46.1	50.4	56.3	61.5	66.3	67.7	66.6	63.8	59.2	53.4	49.2	(31)
		MCDB	49.3	49.1	56.0	66.3	70.6	75.8	78.8	76.8	71.6	63.7	56.6	51.3	(32)
	10%	WB	44.8	43.8	48.3	54.5	59.2	64.0	66.2	64.6	61.8	57.4	51.2	46.8	(33)
		MCDB	46.3	46.2	53.4	63.1	67.7	72.2	76.9	72.8	68.0	61.1	53.9	48.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	9.9	14.3	18.0	17.7	17.8	18.5	17.5	16.4	13.4	9.7	8.0	(35)
		MCDBR	9.2	13.8	21.1	25.3	23.1	24.7	26.3	24.7	22.9	17.7	11.7	9.5	(36)
	5% WB	MCWBR	8.0	9.9	12.3	12.3	10.8	11.2	9.4	10.5	11.2	10.8	8.8	7.9	(37)
		MCWBR	8.7	11.8	18.5	21.7	20.6	20.4	21.9	20.7	20.1	15.9	11.1	9.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.324	0.346	0.376	0.410	0.413	0.412	0.424	0.417	0.389	0.360	0.341	0.318	(40)
		taud	2.369	2.344	2.273	2.219	2.245	2.268	2.244	2.284	2.351	2.415	2.382	2.353	(41)
	Ebn at Noon		213	242	256	261	266	267	261	256	250	234	203	192	(42)
		Edh at Noon	19	26	34	40	41	41	41	37	31	24	19	17	(43)
(44) All-Sky Solar Radiation	RadAvg		217	431	784	1272	1548	1651	1579	1331	968	565	259	168	(44)
	RadStd		27	75	107	141	144	132	202	114	115	54	30	19	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (67 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page